



An illustration of where a firewall would be located in a network

threats, take appropriate actions when necessary and granularly modify the firewall configuration.

Most firewalls search their rules starting from the first through the last. When the firewall finds a rule that matches the packet being examined, the firewall honours it, does what the rule says and looks no further. For example, if the firewall finds the general rule allowing any website connections first, it honours this rule and doesn't look further for rules that might deny such a connection. So the order of firewall rules is a very important factor for an administrator to decide the right access policy.

The different types of firewalls are described below:

Software firewall. Software firewall is specialised software running on a personal computer and needs no extra hardware device. These firewalls are low-priced and sometimes free. But software firewalls are suitable only for small applications such as protecting a single personal computer or a small home network. Moreover, due to limited detection capabilities, these firewalls are prone to a variety of attacks.

Hardware firewall. Hardware firewalls employ dedicated devices designed to protect large networks. These can refer to minor firewall capabilities such as network address translation, built into a broadband router, but more often they refer to specialised firewall devices that provide intelligent filtering and logging services as well as NAT and basic antivirus capabilities. Hardware firewalls are very secure and generally provide more protec-

tion than software firewalls.

Packet filtering firewall. Firewalls based on packet filtering mechanism examine the IP header information carrying source and destination addresses, and ports in each inbound

and outbound traffic. Packet filtering firewall usually works at the third layer (network layer) of open system interconnection (OSI) model. It takes decision whether a packet from some specific IP address should be allowed.

However, packet filtering may be worthless if the attacker takes advantage of IP spoofing, which refers to the practice of forging the source IP address in a packet. In this way, such type of firewall makes no sense of protection and allows traffic if source and destination IP addresses of packets correspond to allowed IP addresses in the access policy of the firewall.

In packet filtering firewalls, port information may also be used to block idle or non-standard ports such as a listening port opened by a Trojan horse. In this way, these firewalls block all the inbound and outbound traffic through specified ports. Very often, hackers use port scanner software to find any open ports in a computer and launch attacks. Such firewalls block the packet sent from the hacker to the Trojan horse listening port. It is worth mentioning here that packet filtering firewall cannot be used for content-based filtering and e-mail attachments.

Application filtering firewall. Application filtering firewall works at the seventh layer (application layer) of OSI model to examine the traffic and implements additional security and access control services. This firewall is implemented as hosts running proxy servers. The proxy server does not allow a direct connection between two networks and instead accepts requests and executes them on behalf of the user. For example, if the user

is on a network behind a proxy firewall in www.xyz.com, the request goes to the proxy firewall, which gets the page on behalf of the user and passes it to the user.

Proxy servers are capable of examining traffic on the basis of source IP address, destination IP address, port information and application information. For instance, administrators can set the firewall to block all the incoming packets belonging to abc.exe files, which are often infected by cyber attacks. Additionally, proxy servers are capable of event logging, auditing and reporting. Thus application filtering firewalls are more versatile and enforce tighter security policies.

Antivirus and antispyware software

Antivirus software is a very important tool to safeguard a computer system against cyber attacks. It examines the contents of each file and tries to find some specific patterns of some bits—called virus signature—of something known to be harmful.

If antivirus software finds some virus in the file, it gives a warning that the particular file is infected with the virus and provides several options to the user to take decision like disinfecting the infected file, removing the file or keeping it intact. In this way, this method protects users only from already known viruses. As soon as a new virus attack is learnt by the antivirus vendors, an updated set of virus signature is added as patch files to the antivirus software.

Some antivirus software use a heuristic algorithm to find viruses based on common behaviours, enabling them to detect and eliminate novel as well as already known viruses. Antispyware software is a specialised kind of software that helps to track down spyware and block them. Some adware eliminators are also available that block adware and block annoying pop-ups. ●

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